

FINAL PROJECT REPORT

Global AI Adoption in Education: A Comprehensive Global Analysis

Data Analytics with Tableau | Prepared by Jeet Pal

1. Introduction

1.1. Project overviews

This project analyzes AI adoption in education across countries and regions for 2015–2026. It examines student AI usage, teacher AI usage, school AI adoption, regional differences, popular AI tools, urban–rural usage, curriculum integration, and government AI policy. Tableau is used to create interactive worksheets, dashboards, and a seven-scene analytical Story.

1.2. Objectives

- Study the overall AI adoption trend from 2015–2026.
- Compare student and teacher AI usage.
- Study urban-rural differences and regional adoption.
- Identify commonly occurring AI tools.
- Explore internet access, policy and curriculum factors.
- Analyse gender gap and school adoption by region.
- Present the results through Tableau dashboards, story points and Tableau Public.
- Integrate the published view into a Flask/Bootstrap web page.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

AI adoption in education is not uniform across countries, regions, stakeholder groups, or access contexts. The project addresses the need to organize adoption indicators into an interactive analytical solution that makes global patterns, disparities, trends, and stakeholder differences easy to understand.

2.2. Project Proposal (Proposed Solution)

The proposed solution is a Tableau analytics project supported by an interactive Story. It uses KPIs, geographic views, comparative charts, trend analysis, categorical views, and Country, Region, and Year filters. The completed Story covers the global challenge, geographic disparity, temporal trends, tool competition, digital equity, stakeholder engagement, and conclusions.

2.3. Initial Project Planning

The planned workflow was: understand the dataset; explore and preprocess the data; build focused Tableau worksheets; assemble dashboards and Story scenes; publish the Story to Tableau Public; build a Flask wrapper; embed the public Tableau Story; and deploy the Flask application publicly.

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

The project uses the Global AI in Education dataset supplied for the project analysis. The working dataset contains 1,360 rows and 16 columns and is approximately 0.35 MB. The analysis uses country, region, year, student usage, teacher usage, school adoption, AI tools, urban/rural usage, curriculum, and government-policy related fields available in the project dataset.

3.2. Data Quality Report

Dataset size: 1,360 records and 16 columns; approximate size: 0.35 MB; time coverage: 2015–2026. Country, Region, and Year are used for interactive analysis. Adoption percentage measures support KPI and comparative views. The project includes an Adoption Categories calculated field for Low (0–33%), Moderate (33–66%), and High (66–100%).

3.3. Data Exploration and Preprocessing

The data was explored so the relevant geographic, time, stakeholder, and adoption fields could be used consistently in Tableau. Measures required for KPI and comparison views were prepared, and the Adoption Categories calculated field was created for categorical analysis.

4. Data Visualization

4.1. Framing Business Questions

Key questions include: How does AI adoption differ across regions and countries? How has adoption changed from 2015 to 2026? How do student, teacher, and school adoption compare? Which AI tools have the strongest representation? Is there an urban–rural divide? How are curriculum and policy represented? Can Country, Region, and Year filters support focused exploration?

4.2. Developing Visualizations

The project integrates nine visualization types identified in the project specification: KPI Cards, Geospatial Map, Bar Chart, Grouped Bar Chart, Area/Trend Line Chart, Pie Chart, Waterfall Chart, Donut Chart, and Bubble/Categorical Chart. The final trend chart contains Student AI Usage %, Teacher AI Usage %, and School AI Adoption % by Year.

5. Dashboard

5.1. Dashboard Design File

The main Global AI Adoption in Education dashboard consolidates key indicators and visualizations. Supporting views include regional adoption, global adoption mapping, student usage by country, popular AI tools, urban versus rural usage, curriculum integration, government AI policy, and student-versus-teacher usage. Country, Region, and Year filters support interactive exploration.

6. Report

6.1. Story Design File

The Tableau Story contains seven scenes: (1) Introduction — Global AI in Education Challenge; (2) Geographic Disparity — Regional AI Adoption Variations; (3) Temporal Trends — AI Adoption Journey (2015–2026); (4) Tool Competition — Leading AI Platforms Market Share; (5) Digital Equity — Urban-Rural Adoption Divide; (6) Stakeholder Engagement — Student vs Teacher Adoption Parity; (7) Conclusions and Recommendations — Strategic Pathways Forward.

7. Performance Testing

7.1 Utilization of Data filters

Three primary interactive filters are used: Country, Region, and Year.

7.2 No of Calculation Field

The project uses the Adoption Categories calculated field, grouping adoption values into Low (0–33%), Moderate (33–66%), and High (66–100%).

7.3 No of Visualization

Nine visualization types are integrated: KPI Cards, Geospatial Map, Bar Chart, Grouped Bar Chart, Area/Trend Line Chart, Pie Chart, Waterfall Chart, Donut Chart, and Bubble/Categorical Chart.

8. Conclusion/Observation

The completed analysis provides a consolidated view of AI adoption in education across geography, time, stakeholders, tools, and digital-equity dimensions. Regional comparisons reveal differences that can be hidden by a single global average. The 2015–2026 trend view provides a longitudinal perspective, while

student-versus-teacher and urban-versus-rural views add stakeholder and equity context. The Tableau Story turns these views into a structured narrative for interpretation and recommendations.

9. Future Scope

Future work can include regularly refreshed data beyond 2026, institution-level segmentation, infrastructure/connectivity and AI-training indicators, forecasting, additional policy/governance variables, and validated external datasets for richer cross-country comparisons.

10. Appendix

10.1. Source Code(if any)

The web-integration source code is maintained in the public GitHub repository. It contains app.py, templates/index.html, requirements.txt, and .gitignore. The Flask page embeds the published Tableau Story.

10.2. GitHub & Project Demo Link

GitHub Repository: https://github.com/jeetpal-tech/GlobalAI_WebIntegration

Project Demo / Flask Web Integration: <https://globalai-webintegration.onrender.com>

Tableau Public Story:

https://public.tableau.com/app/profile/jeet.pal7266/viz/GlobalAIAdoptioninEducation_/Story1?publish=yes